MILITARY SPECIFICATION

DETONATOR, ELECTRIC, MARK 71 MOD 1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 71 MOD 1 Electric Detonator, referred to herein as the detonator.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-1876	Aluminum Foil.
UU-T-81	Tags, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1375

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PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-676	Boxes, Setup.

MILITARY

JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier Materials, Waterproof, Flexible.
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.

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MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND (Code Ident 10001)

LD 267058	Test Set, MARK 152 MOD 0.
LD 486196	Jolt and Vibration Fixture.
LD 486197	Drop Test Fixture.

NAVAL AIR SYSTEMS COMMAND (Code Ident 30003)

DL 3024245	Detonator, Electric, MARK 71 MOD 1.

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND (Code Ident 10001)

OD-6821	Test Set, MARK 152 MOD 0.

(Copies of drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test Methods for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Item description. The detonator contains sensitive explosives which are electrically initiated by the dissipation of energy through a bridgewire. The detonator initiates the explosive train of a safety-arming (S-A) device or fuze of a guided missile. The completely assembled detonator is shown on Drawing 3024245. The major components and assemblies of the detonator are as follows (see DL 3024245):

a. Sleeve (see Drawing 3024249).

b. Cup (see Drawing 3024246).

c. Bridged plug assembly (see Drawing 3024250).

d. Ignition assembly (see Drawing 3024247).

e. Plug and lead wire assembly (see Drawing 3024248).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Electrical.

3.2.1.1.1 Bridgewire resistance. Prior to shunting the lead wires, the bridgewire resistance between the two lead wires shall be 5.0 ±2.0 ohms (see 4.8.1.1).

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3.2.1.1.2 Insulation resistance. After shunting the lead wires, the insulation resistance between the lead wires and the cup shall be 50 megohms (Mohms) minimum when measured under an applied potential of 750 ±75 volts alternating current root-mean-square (Vac rms) (see 4.8.1.2).

3.2.1.2 Function and output. The detonator shall function on the first and only try and the output shall produce an average corrected depth of dent of 0.0125 inch minimum when initiated in accordance with 4.8.2. The depth of any single dent, or the average corrected depth of dent less three standard deviations, shall be 0.010 inch minimum.

3.2.2 Mechanical.

3.2.2.1 Lead wire flexibility. Each lead wire shall be capable of withstanding, without breaking, one set of successive bends in opposite directions of 90, 180, and 90 degrees (see 4.8.3.1).

3.2.2.2 Radiographic. The detonator shall meet the following requirements when radiographically inspected (see 4.8.3.2):

a. There shall be no evidence of lead wires contacting the cup or each other within the plug (see Drawing 3024245).

b. The pentaerythritol tetranitrate (PETN) and the two lead azide charges shall be in accordance with Drawing 3024245, and shall be identifiable and without voids.

c. The lead styphnate flash charge shall be in contact with the lead azide charge (see Drawing 3024245).

d. The ignition charge shall be in contact with the lead styphnate flash charge (see Drawing 3024245).

e. Cup thickness, plug seating, and wire crimping shall be in accordance with the applicable drawings.

f. There shall be no evidence of metal slivers, wire clippings, voids, or other foreign material.

3.2.2.3 Visual. The detonator shall meet the following requirements when visually inspected (see 4.8.3.3):

a. There shall be no evidence of external damage, such as buckles, cracks, dents, or other blemishes, which may interfere with functioning or fit at the next assembly.

b. The detonator shall be crimped 360 degrees around the plug and shall have a uniform seal around the crimp (see Drawing 3024245).

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c. There shall be no evidence of kinks, nicks, or scratches on the insulated portion of the lead wires.

d. The lead wires shall be positively shorted together (see Drawing 3024245).

3.2.3 Environmental. The detonator shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the following environments.

3.2.3.1 Transportation vibration - temperature. The detonator shall not deflagrate, initiate, break, or release any explosive material during exposure to the environment of 4.8.4.1. The detonator shall be safe (see 6.3) and shall meet the requirements of 3.2.1.2 after exposure.

3.2.3.2 Thermal shock. The detonator shall not deflagrate, initiate, break, or release any explosive material during exposure to the environment of 4.8.4.2. The detonator shall be safe (see 6.3) and shall meet the requirements of 3.2.1.2 after exposure.

3.2.3.3 Temperature and humidity. The detonator shall not deflagrate, initiate, break, or release any explosive material during exposure to the environment of 4.8.4.3. The detonator shall be safe (see 6.3) and shall meet the requirements of 3.2.1.2 after exposure.

3.2.3.4 Aircraft vibration. The detonator shall not deflagrate, initiate, break, or release any explosive material during exposure to the environment of 4.8.4.4. The detonator shall be safe (see 6.3) and shall meet the requirements of 3.2.1.2 after exposure.

3.2.4 Safety.

3.2.4.1 Bridgewire safety. The detonator shall not initiate when discharge energy from a charged capacitor is applied to the bridgewire (see 4.8.5.1).

3.2.4.2 Jolt. The detonator shall not initiate nor release any explosive material, and no explosive elements shall initiate during and after exposure to the environment of 4.8.5.2. The detonator shall be safe to handle and dispose of (see 6.3) after exposure.

3.2.4.3 Forty-foot drop. The detonator shall not initiate nor release any explosive material, and no explosive elements shall initiate during and after exposure to the environment of 4.8.5.3. The detonator shall be safe to handle and dispose of (see 6.3) after exposure.

3.3 First article. The contractor shall furnish 300 detonators for first article inspection and approval for each contract or purchase order (see 4.3 and 6.2.3).

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3.4 Design and construction.

3.4.1 Production drawings. The detonator shall meet all the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on DL 3024245.

3.4.2 Standards of manufacture.

3.4.2.1 Assembly conditions. The detonator shall be assembled in a careful, neat, and orderly manner to prevent damage to parts and inclusion of moisture, dust, or other foreign material.

3.4.2.2 Working environment. Unless otherwise specified herein or on the drawings, the working environment shall meet the minimum levels and requirements specified in MIL-STD-1695.

3.4.2.3 Materials. Materials used in the manufacture of the detonator shall be as specified on the drawings and herein. The contractor shall not change material suppliers during a specified contract unless authorized by the Government.

3.4.2.4 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specifed herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2. Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspections (see 4.3).

b. Quality conformance inspections (see 4.4).

4.3 First article. Unless otherwise specified in the contract or purchase order, first article inspections shall be performed by the contractor, after award of contract and prior to production. First article inspections shall be performed on 300 sample detonators which have been produced

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with equipment, material, and procedures to be used in production (see 3.3). First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

4.3.1 First article inspection sequence. The first article of 300 detonators shall be subjected to the inspections of Table I in the sequence shown. After receipt of first article approval, first article detonators shall be disposed of at the discretion of the cognizant demolition officer.

4.3.2 First article acceptance and rejection criteria. The first article shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Table I, the applicable performance requirements of Section 3 shall be met on the first and only trial; one or more failures shall be cause for refusal to grant first article approval.

4.4 Quality conformance inspections.

4.4.1 Inspection lot . An inspection lot, as defined in MIL-STD-105, shall consist of completed detonators fabricated under the same conditions and offered for inspection at the same time. In addition, an inspection lot shall include:

a. Detonators bearing the same inspection lot number.

b. Normal lead styphnate, milled and prepared as specified on Drawing 3024245, from one batch or a blending of batches, resulting in homogeneity of all lead styphnate used in loading the entire lot.

c. Lead azide from one batch or a blending of batches, resulting in homogeneity of all lead azide used in loading the entire lot.

d. PETN from one batch or a blending of batches, resulting in homogeneity of all PETN used in loading the entire lot.

4.4.1.1 Inspection lot size. Unless otherwise specified in the contract or purchase order, an inspection lot shall consist of not less than 1,000 nor more than 3,000 detonators, including detonators required for destructive tests (see 6.2.1).

4.4.1.2 Inspection lot inspection sequence. After the inspection lot has been formed and identified by lot number, 150 detonators shall be randomly sampled from the lot in accordance with MIL-STD-105 and subjected to the inspections of Table II in the sequence shown. Detonators subjected to the inspections of Table II shall be considered destroyed, labeled appropriately, and disposed of at the discretion of the cognizant demolition officer.

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TABLE I. First article inspections.

Inspection	Number of first article detonators	Requirement paragraph	Method paragraph
Group I			
Drawing conformance	All first article detonators (300 total)	3.4.1	4.8.6
Visual		3.2.2.3	4.8.3.3
Packaging		5.	4.8.7
Insulation resistance		3.2.1.1.2	4.8.1.2
Bridgewire resistance		3.2.1.1.1	4.8.1.1
Radiographic		3.2.2.2	4.8.3.2
Group II			
Bridgewire safety	30 from Group I	3.2.4.1	4.8.5.1
Bridgewire resistance[1/]		3.2.1.1.1	4.8.1.1
Group III			
Lead wire flexibility	15 from Group I	3.2.2.1	4.8.3.1
Jolt		3.2.4.2	4.8.5.2
Group IV			
Forty-foot drop	15 from Group I	3.2.4.3	4.8.5.3
Group V			
Bridgewire resistance[2/]	Groups III and IV (30 total)	3.2.1.1.1	4.8.1.1
Group VI			
Transportation vibration-temperature	240 from Group I	3.2.3.1	4.8.4.1
Group VII			
Temperature and humidity	120 from Group VI	3.2.3.3	4.8.4.3
Group VIII			
Thermal shock	120 from Group VI	3.2.3.2	4.8.4.2
Group IX			
Aircraft vibration	Groups VII and VIII (240 total)	3.2.3.4	4.8.4.4
Bridgewire resistance		3.2.1.1.1	4.8.1.1
Insulation resistance		3.2.1.1.2	4.8.1.2
Function and output[3/]		3.2.1.2	4.8.2
a. 80 at -65 ±5°F			
b. 80 at ambient temp			
c. 80 at +165 ±5°F			

[1/] After test, fire for information only.

[2/] This test for information only.

[3/] Separate each group of 120 (120 from Group VII and 120 from Group VIII) into three subgroups of 40 each; test one subgroup at -65°F, one subgroup at ambient, and one subgroup at +165°F.

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TABLE II. Inspection lot sample inspections.

Inspection	Number of sample detonators	Requirement paragraph	Method paragraph	Acceptance criteria/ disposition of defectives
Group I				
Drawing conformance	All sample detonators (150 total)	3.4.1	4.8.6	Acceptance criteria and disposition of defectives shall be as specified on the drawings.
Visual		3.2.2.3	4.8.3.3	1 failure = screen entire lot and correct defectives.
Packaging		5.	4.8.7	
Bridgewire resistance		3.2.1.1.1	4.8.1.1	1 failure = screen entire lot and scrap defectives.
Insulation resistance		3.2.1.1.2	4.8.1.2	
Radiographic		3.2.2.2	4.8.3.2	
Group II				
Bridgewire safety	15 from Group I	3.2.4.1	4.8.5.1	1 failure = reject lot.
Bridgewire resistance[1/]		3.2.1.1.1	4.8.1.1	N/A, for information only.
Group III				1 failure = reject lot.
Lead wire flexibility	15 from Group I	3.2.2.1	4.8.3.2	
Jolt		3.2.4.2	4.8.5.2	
Group IV				
Forty-foot drop	15 from Group I	3.2.4.3	4.8.5.3	
Group V				
Bridgewire resistance[1/]	Groups III and IV (30 total)	3.2.1.1.1	4.8.1.1	N/A, for information only.

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TABLE II. Inspection lot sample inspections (continued).

Inspection	Number of sample detonators	Requirement paragraph	Method paragraph	Acceptance criteria/ disposition of defectives
Group VI Transportation vibration-temperature	105 from Group I	3.2.3.1	4.8.4.1	1 failure = reject lot.
Group VII Temperature and humidity	54 from Group VI	3.2.3.3	4.8.4.3	
Group VIII Thermal shock	51 from Group VI	3.2.3.2	4.8.4.2	
Group IX				
Aircraft vibration	Groups VII and VIII (105 total)	3.2.3.4	4.8.4.4	
Bridgewire resistance		3.2.1.1.1	4.8.1.1	
Insulation resistance		3.2.1.1.2	4.8.1.2	
Function and output[2/] a. 35 at -65 ±5°F b. 35 at ambient temp c. 35 at +165 ±5°F		3.2.1.2	4.8.2	

1/ After test, fire for information only.

2/ Separate Group VII into three subgroups of 18 each; test one subgroup at -65°F, one subgroup at ambient, and one subgroup at +165°F. Separate Group VIII into three subgroups of 17 each; test one subgroup at -65°F, one subgroup at ambient, and one subgroup at +165°F.

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4.4.1.3 Inspection lot acceptance and rejection criteria. The detonator shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Table II, acceptance or rejection shall be as specified in the table.

4.5 Test equipment. All test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order, the contractor shall prepare a test equipment calibration plan in accordance with the Data Item Description (DID) cited in 6.2.2 and which meets the requirements of MIL-STD-45662. The calibration plan shall be accepted by the Government prior to use.

4.5.1 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions and to the accuracies specifed on the drawings and herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. Any error introduced by the test equipment shall not be applicable toward widening the allowable tolerance of the requirement being measured. All test equipment shall be calibrated and maintained as specified in 4.5.

4.5.2 Special test equipment and circuits. When special conditions of test are not imposed, commerically-available, direct-measurement equipment may be employed. When special test equipment and circuits, other than as specified herein, are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall conform to 4.5 and 4.5.1 and are subject to approval by the Government.

4.5.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance to the dimensional requirements of this specification (see 4.8.2.3.a) and the applicable drawings.

4.6 Test environment. Unless otherwise specified herein or on the drawings, the inspection methods of 4.8 shall be performed under the environmental conditions specified in 3.4.2.2.

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4.7 Test conditions and provisions. Unless otherwise specified herein, the following test conditions and provisions shall apply during performance of the inspection methods of 4.8:

a. When specified in the contract or purchase order, the contractor shall prepare a test plan and test procedures in accordance with MIL-T-18303 and the DID cited in 6.2.2. The test plan and test procedures shall be accepted by the Government prior to use.

b. When specified in the contract or purchase order, the contractor shall prepare a test report in accordance with MIL-STD-831 and the DID cited in 6.2.2 for each lot produced under a specified contract, including first article, inspection, and rejected lots.

c. When specified in the contract or purchase order, the contractor shall prepare an inspection record in accordance with the DID cited in 6.2.2 for the first article (see 4.3) and for each inspection lot (see 4.4) produced under a specified contract. The results of all inspections specified herein shall be recorded on the inspection record and may be extracted from the inspection and acceptance records maintained by the contractor.

d. The contractor shall not deviate from the inspection methods of 4.8 or the test plan and test procedures of a. above unless authorized by the Government.

e. The inspection methods of 4.8 shall be performed in the sequences shown in Tables I and II, as applicable. When feasible, sequential tests may be combined provided it can be demonstrated by the contractor that the applicable requirements of Sections 3 and 5 have been individually met. Additional tests shall not be added unless authorized by the Government.

4.8 Inspection methods. To determine conformance to the requirements of Sections 3 and 5, the following inspections shall be performed in accordance with the test conditions and provisions of 4.7, and in the sequences shown in Tables I and II, as applicable.

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WARNING

The detonator contains explosive components which are sensitive and hazardous. Proper safety precautions shall be observed during handling and testing of the detonator (see 6.4).

4.8.1 Electrical tests.

4.8.1.1 Bridgewire resistance. To determine conformance to 3.2.1.1.1, test the detonator prior to shunting the lead wires as follows:

WARNING

The bridgewire resistance test involves hazardous operations. Proper safety precautions shall be observed (see 6.4). The test shall be performed with the detonator installed in a safety chamber capable of containing an output of 260 milligrams (mg) of explosive material.

a. Place the detonator in a safety chamber, connect the required electrical monitoring equipment, and close the chamber door.

b. Measure the resistance between the two lead wires in accordance with MIL-STD-202, Method 303.

4.8.1.2 Insulation resistance. To determine conformance to 3.2.1.1.2, test the detonator after shunting the lead wires as follows:

WARNING

The insulation resistance test involves hazardous operations. Proper safety precautions shall be observed (see 6.4). The test shall be performed with the detonator installed in a safety chamber capable of containing an output of 260 mg of explosive material.

a. Place the detonator in a safety chamber. Connect the required electrical monitoring equipment using the shunted, twisted lead wires and the detonator cup as terminals. Close the chamber door.

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CAUTION

At no time shall the test current through the detonator exceed 5 milliamperes (mA). Current in excess of 5 mA may cause initiation of the explosive components of the detonator.

b. Apply a potential of 750 ±75 Vac rms for 1 second minimum to the detonator.

c. Measure the resistance between the lead wires and the detonator cup in accordance with MIL-STD-202, Method 302.

4.8.2 Function and output. To determine conformance to 3.2.1.2, measure the hardness of the dent block at the required test temperature (see 4.8.2.3.b), temperature-condition the dent block and the detonator as specified in 4.8.2.1, test the detonator as specified in 4.8.2.2, and measure the depth of dent as specified in 4.8.2.3.

4.8.2.1 Temperature-conditioning for function and output test. Prior to test, the detonator shall be mounted in a polystyrene confining sleeve. The dent block (see 4.8.2.3) and the sleeve containing the detonator shall then be temperature-conditioned as specified below to the test temperature specified for the individual test of Table I or II. The detonator shall be subjected to the test of 4.8.2.2 within 1 minute maximum after temperature-conditioning or removal from the temperature chamber.

a. Room ambient temperature. Place the dent block and the sleeve containing the detonator in a suitable, safe location (see 6.4) at room ambient temperature for 60 minutes minimum prior to test.

b. High temperature. Place the dent block and the sleeve containing the detonator into a temperature chamber which has been preconditioned to +165 ±5 degrees Fahrenheit (°F). Maintain the temperature for 60 minutes minimum prior to test.

c. Low temperature. Place the dent block and the sleeve containing the detonator into a temperature chamber which has been preconditioned to -65 ±5°F. Maintain the temperature for 60 minutes minimum prior to test.

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4.8.2.2 Function and output test procedure. After temperature-conditioning as specified in 4.8.2.1, test the detonator as follows:

WARNING

The function and output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 6.4). The detonator shall be initiated in a safety chamber capable of containing an output of 260 mg of explosive material.

a. Place the detonator and the dent block of 4.8.2.3 into a safety chamber.

b. Initiate the detonator in accordance with the procedures of OD-6821, using a MARK 152 MOD 0 Test Set (see LD 267508), except the firing capacitor shall be 1.00 ±0.01 microfarad (μF) charged to 38.5 ±0.1 volts direct current (Vdc), and shall discharge through a 4.0 ±0.2 ohm resistor in series with the detonator bridgewire. The firing switch shall close without chatter or contact bounce.

c. After the test, remove the detonator and dent block from the safety chamber. Remove all foreign material from the dent block and measure the depth of dent as specified in 4.8.2.3.

4.8.2.3 Dent block, and dent measurement equipment and procedure. The dent block, and dent measurement equipment and procedure shall be in accordance with MIL-STD-331, Test 301, except:

a. The dial indicator height gage shall be graduated in 0.0005 inch units and shall be accurate to 0.0005 inch minimum.

b. The hardness of all dent blocks used in a series of tests shall be measured after temperature-conditioning to the required test temperature and shall not vary more than 10 points, Rockwell B, in accordance with ASTM E-18. The hardness of each dent block shall be determined at two places, each 0.125 to 0.250 inch from the saw-cut edge on the centerline of the test surface. The dent block hardness shall be recorded on the inspection record (see 4.7.c).

NOTE: Prior to hardness measurements, dent blocks to be used in a series of tests at a specified temperature shall be temperature-conditioned to that temperature as specified

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in 4.8.2.1. After hardness measurements and prior to test, dent blocks shall again be temperature-conditioned along with the detonators as specified in 4.8.2.1.

c. The average depth of dent of all blocks in a series of tests shall be adjusted by the formula:

$$d_c = d_o + 1/3\ (H - 83)$$

where

d_c = the corrected average depth of dent in 1/1000 inch (corrected for dent block hardness).

d_o = the observed (measured in accordance with MIL-STD-331, Test 301) average depth in 1/1000 inch of all blocks in a series of tests.

H = the average hardness, Rockwell B, of all the dent blocks in a series of tests.

d. The depth of each single dent shall be measured and recorded on the inspection record (see 4.7.c). The average hardness (H), average depth of dent (d_o), and the corrected average depth of dent (d_c) shall be calculated using the formula of c. above, standard deviation and 3-sigma low limit (corrected average depth less three standard deviations).

4.8.3 Mechanical tests.

4.8.3.1 Lead wire flexibility. To determine conformance to 3.2.2.1, subject each lead wire, separately, to successive bends in opposite directions of 90, 180, and 90 degrees while applying a tensile force of 135 ±14 grams to the wire. Bending shall occur as close as possible to the plug face in a plane normal to the plug face, using the point of exit as a fulcrum, at a rate of 3 seconds nominal per bend.

4.8.3.2 Radiographic. To determine conformance to 3.2.2.2, subject the detonator to radiographic inspection along the longitudinal axis (see Drawing 3024245) in accordance with MIL-STD-202, Method 209.

4.8.3.3 Visual. To determine conformance to 3.2.2.3 and the general requirements of Section 3, visually inspect the detonator.

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4.8.4 Environmental tests.

4.8.4.1 Transportation vibration - temperature. To determine conformance to 3.2.3.1, test the detonator as follows:

a. Mount the detonator to the test fixture of LD 486196. To prevent breakage during the test, tape the lead wires flat against the test fixture.

b. Mount the test fixture to a vibration machine and tighten the mounting bolts to a torque of 23 ±3 foot-pounds (ft-lb).

c. Subject the detonator to the test of MIL-STD-331, Test 119, Procedure 2.

d. After the test, examine the detonator for any evidence of deflagration, initiation, breakage, and release of explosive material.

4.8.4.2 Thermal shock. To determine conformance to 3.2.3.2, subject the detonator to the test of MIL-STD-202, Method 107, Test Condition A, except Step 3 shall be +165 ±5°F. After the test, examine the detonator for any evidence of deflagration, initiation, breakage, and release of explosive material.

4.8.4.3 Temperature and humidity. To determine conformance to 3.2.3.3, subject the detonator to the test of MIL-STD-331, Test 105, procedure optional. After the test, examine the detonator for any evidence of deflagration, initiation, breakage, and release of explosive material.

4.8.4.4 Aircraft vibration. To determine conformance to 3.2.3.4, test the detonator as follows:

a. Repeat 4.8.4.1.a and b.

b. Subject the detonator to random noise vibration in accordance with the procedures of MIL-STD-810, Method 514, at the levels specified on Figure 1 and applied along each of two orthogonal axes (one of which passes through the cylindrical axis (see Drawing 3024245)) for a period of 4.5 hours minimum per axis (total of 9 hours minimum per detonator).

c. After the test, examine the detonator for any evidence of deflagration, initiation, breakage, and release of explosive material.

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4.8.5 Safety tests.

4.8.5.1 Bridgewire safety. To determine conformance to 3.2.4.1, test the detonator as follows:

WARNING

The bridgewire safety test involves hazardous operations. Proper safety precautions shall be observed (see 6.4). The test shall be performed with the detonator installed in a safety chamber capable of containing an output of 260 mg of explosive material.

a. Install the detonator in a safety chamber. Connect the firing circuit of 4.8.2.2.b, except the firing capacitor shall be charged to 18.0 +0.0, -0.5 Vdc and the 4-ohm resistor shall be shorted-out. Close the chamber door.

b. Apply a fire pulse to the detonator.

c. Open the chamber door and examine the detonator to determine conformance to 3.2.4.1.

NOTE: Step d. below is optional, but is recommended to assure safety during subsequent disposal of the detonator.

d. Close the chamber door. Using the firing circuit of a. above, apply 38.5 ±0.1 Vdc to initiate the detonator.

4.8.5.2 Jolt. To determine conformance to 3.2.4.2, test the detonator as follows:

a. Mount 15 detonators to the test fixture of LD 486196. To prevent breakage during the test, tape the lead wires flat against the test fixture.

b. Mount the test fixture to a jolt machine and tighten the mounting bolts to a torque of 23 ±3 ft-lb.

c. Subject the detonators to the test of MIL-STD-331, Test 101, Procedure II.

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d. After the test, examine the detonator for any evidence of initiation and release of explosive material.

4.8.5.3 Forty-foot drop. To determine conformance to 3.2.4.3, test the detonator as follows:

a. Mount 15 detonators to the test fixture of LD 486197. To prevent breakage during the test, tape the lead wires flat against the test fixture.

b. Subject the detonators to the test of MIL-STD-331, Test 103, Procedure 2, except the drop test apparatus shall consist of the following three basic components:

(1) A rigidly supported, hardened steel anvil having a thickness of 2.0 inches minimum.

(2) A carrier to which the test fixture of a. above shall be fastened. The striking surface of the carrier shall have a spherical radius of 6.00 ±0.05 inches, and a cross-sectional diameter of 2.5 inches minimum at the greatest diameter, measured normal to the axis of the carrier. The axis of the carrier and the axis of the striking surface shall coincide within ±0.05 inch.

(3) A vertical guide to guide the carrier during drop to impact with the anvil. To simulate free-fall conditions as close as possible, the surfaces of the guide shall not allow more than a negligible amount of friction to develop between the carrier and the guide during drop.

c. After test, examine the detonator for any evidence of initiation and release of explosive material.

4.8.6 Drawing conformance. To determine conformance to 3.4.1, perform drawing inspections on each batch of explosives and on each lot of parts, subassemblies, and assemblies of the detonator during manufacture, and on each completely assembled detonator after manufacture in accordance with the drawings listed on DL 3024245. Where required during manufacture, the sampling plans and acceptance criteria shall be as specified on the drawings.

4.8.7 Packaging inspection. To determine conformance to Section 5, inspect the packaging, packing, and marking of the detonator prior to shipment.

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5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level A or C, as specified in the contract or purchase order (see 6.2.1).

5.1.1 Level A packaging. Ten detonators shall be bundled together, individually packaged, and tagged as specified on Figure 2.

5.1.2 Level C packaging. Ten detonators shall be bundled together, individually packaged, and tagged as specified on Figure 2 and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.2. Packing. Packing shall be in accordance with FED-STD-102, Level A or C, as specified in the contract or purchase order (see 6.2.1).

5.2.1 Level A packing. Detonators, packaged as specified in 5.1.1, shall be packed in cartons and boxes in accordance with Figure 3.

5.2.2. Level C packing. Detonators, packaged as specified in 5.1.2, shall be packed in cartons and boxes in accordance with Figure 3 and in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior and exterior marking shall be as specified on Figures 2 and 3 and in accordance with MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The detonator is intended for use in a safety-arming device or fuze of a guided missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

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c. Inspection lot size, if other than as specified in 4.4.1.1.

d. Level of preservation-packaging (see 5.1).

e. Level of packing (see 5.2).

f. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements indentified below should be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of the Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data Requirement Title	Applicable DID No.	Options
4.5	Test Equipment Calibration Plan	UDI-T-23742A	------
4.7.a	Test Plan and Test Procedures	DI-T-5204	------
4.7.b	Test Reports	DI-T-4024	------
4.7.c	Inspection Record	DI-T-2072	------

(DIDs related to this specification, and identified in Section 6 should be approved and listed as such in DoD 5000.19L., Vol II, AMSDL. Copies of DIDs required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. The first article should be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a preproduction sample consisting of 300 detonators, as specified in 3.3 and 4.3. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the document's first article.

6.3 Definition of safe. For the environmental requirements specified herein, safe is defined as the ability of the detonator to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life

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which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the detonator to maintain its safety feature in a condition which will permit disposal without injury to personnel.

6.4 Explosives safety precautions. The manufacture and assembly of the detonator requires loading and handling of explosive material which can be hazardous to personnel and equipment. These operations require the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

Preparing activity:
Navy - AS

(Project 1375-N253)

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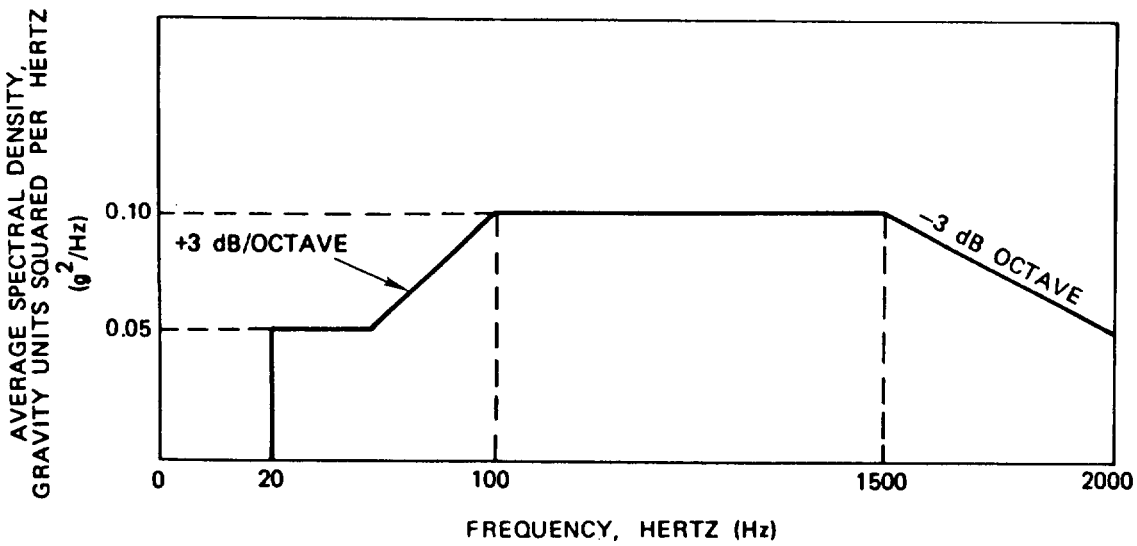
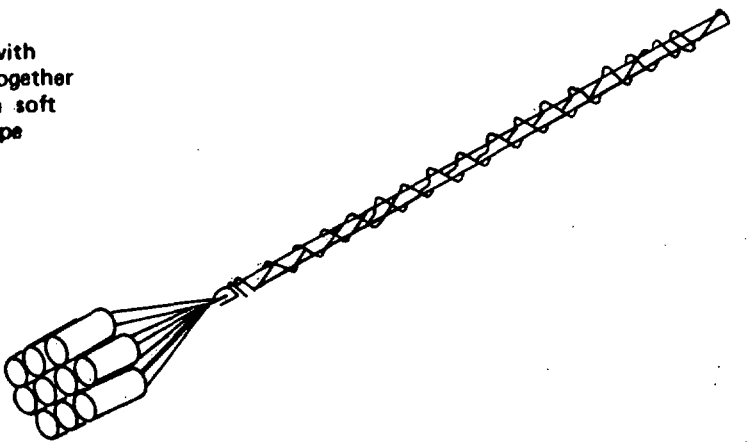


FIGURE 1. Aircraft vibration test levels.

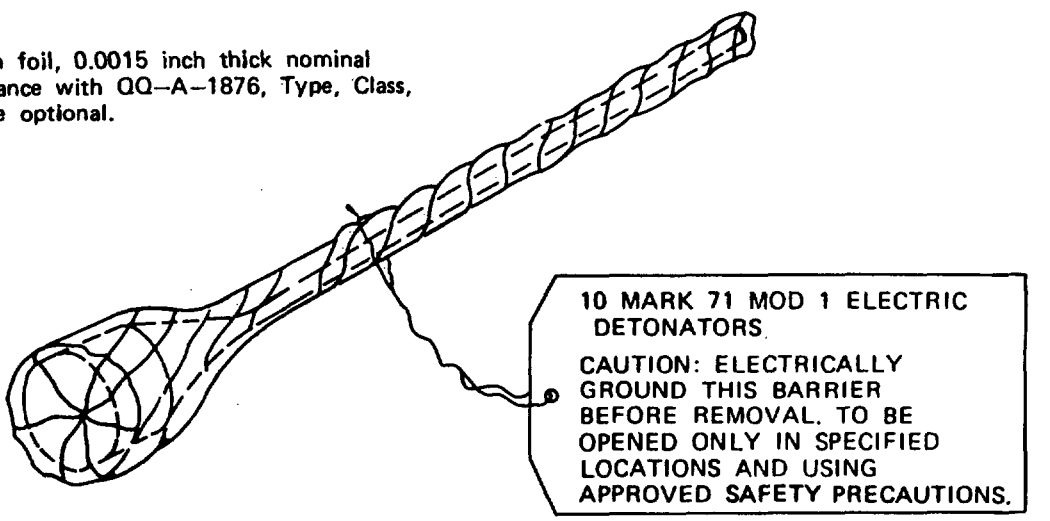
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10 detonators with leads bundled together and bound with soft copper wire (type optional).



Step 1

Aluminum foil, 0.0015 inch thick nominal in accordance with QQ-A-1876, Type, Class, and Grade optional.



Step 2 Tag in accordance with UU-T-81, Type B, Grade 15SU, Size 3 or 4, Color white. Required data may be printed or stamped thereon.

FIGURE 2. Packaging of detonators.

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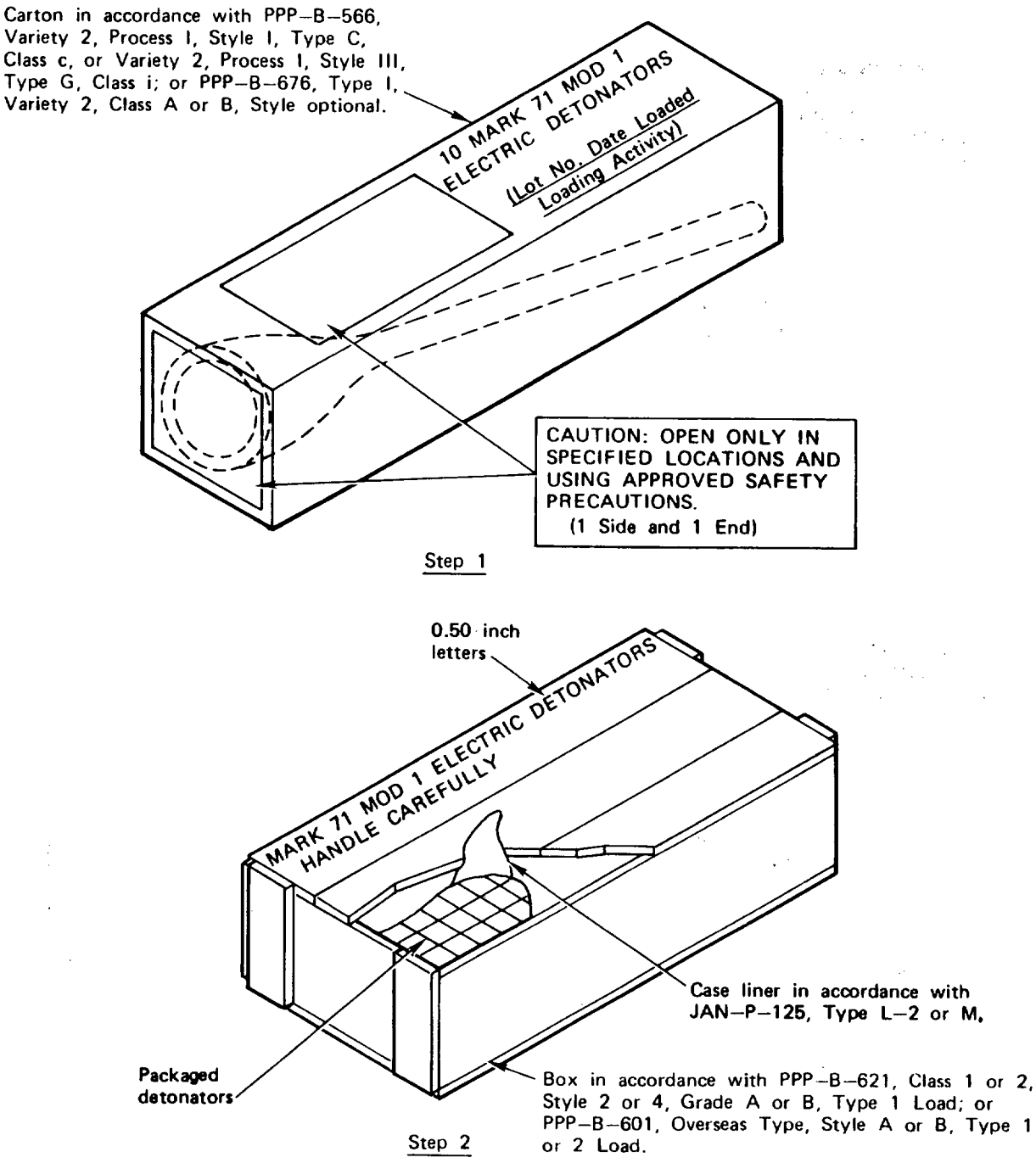


FIGURE 3. Packing of detonators.

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